Powers and products

Evaluate squares and products involving positive and negative values.

Name: _____ Suggested time: 10 minutes

Show your working. Use brackets when substituting negative values.

01 Given a=0, b=0

Evaluate: $a^2 - 4ab + b^2$

02 Given a=1, b=0

Evaluate: $a^2 - 4ab + b^2$

03 Given a=0, b=2

Evaluate: $a^2 - 6ab + b^2$

04 Given a=1, b=4

Evaluate: $a^2 - 2ab + b^2$

05 Given a=4, b=-1

Evaluate: $a^2 - 5ab + b^2$

Answer sheet

Powers and products | 5 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 0

02 1

03 4

04 9

05 37

Worked solutions

Powers and products

01 Given a=0, b=0

$a^2 - 4ab + b^2$

Substitute a=0, b=0.

$(0)^2 - 4(0)(0) + (0)^2$

0

0

02 Given a=1, b=0

$a^2 - 4ab + b^2$

Substitute a=1, b=0.

$(1)^2 - 4(1)(0) + (0)^2$

1

1

03 Given a=0, b=2

$a^2 - 6ab + b^2$

Substitute a=0, b=2.

$(0)^2 - 6(0)(2) + (2)^2$

$0 + 4$

4

04 Given a=1, b=4

$a^2 - 2ab + b^2$

Substitute a=1, b=4.

$(1)^2 - 2(1)(4) + (4)^2$

$1 - 8 + 16$

9

Worked solutions

Powers and products

05 Given a=4, b=-1

$a^2 - 5ab + b^2$

Substitute a=4, b=-1.

$(4)^2 - 5(4)(-1) + (-1)^2$

$16 +20 +1$

37